

DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 6 (AT3)

Case Study

Instructions:

For each question, write the program in the specified language (R or Python), generate the required

visualization, and answer the interpretation sub-questions clearly with justification.

Case Study 1 – Retail Sales Performance Dashboard

Scenario

A retail company has collected one year of sales data from five regions (North, South, East, West, Central). The dataset contains monthly sales, product categories, customer segments, and total revenue.

The management wants to identify:

- The highest-performing region.
- Monthly sales trends.
- Revenue contribution of each product category.
- Relationship between advertising expenditure and sales.
- Sales distribution across regions.

Tasks

1. Select the most suitable visualization for each requirement:
 - Regional revenue comparison
 - Monthly sales trend
 - Product category contribution
 - Advertising expenditure vs sales
 - Regional sales distribution

Most suitable visualization

Requirement	Suitable visualization	Reason
Regional revenue comparison	Bar Plot	Compares revenue across regions clearly
Monthly sales trend	Line Chart	Shows changes and trends over time
Product category contribution	Stacked Bar Chart	Shows each category's contribution to total sales
Advertising expenditure vs sales	Scatter Plot	Shows the relationship/correlation between two numerical variables

Regional sales distribution	Histogram / Density Plot	Shows how sales values are distributed
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2. Using **R**, create:

- Bar Plot
- Stacked Bar Chart
- Histogram
- Density Plot
- Scatter Plot

Sample dataset

Create sample retail sales data

```
sales_data <- data.frame(
  Region = c("North", "South", "East", "West", "Central",
             "North", "South", "East", "West", "Central"),
  Month = c("Jan", "Jan", "Jan", "Jan", "Jan",
            "Feb", "Feb", "Feb", "Feb", "Feb"),
  Category = c("Electronics", "Clothing", "Food", "Electronics", "Clothing",
               "Food", "Electronics", "Clothing", "Food", "Electronics"),
  Sales = c(85000, 70000, 60000, 95000, 75000,
            65000, 90000, 72000, 68000, 98000),
  Advertising = c(12000, 9000, 7000, 15000, 10000,
                  8000, 13000, 9500, 8500, 16000),
  Revenue = c(85000, 70000, 60000, 95000, 75000,
              65000, 90000, 72000, 68000, 98000)
)
```

```
head(sales_data)
```

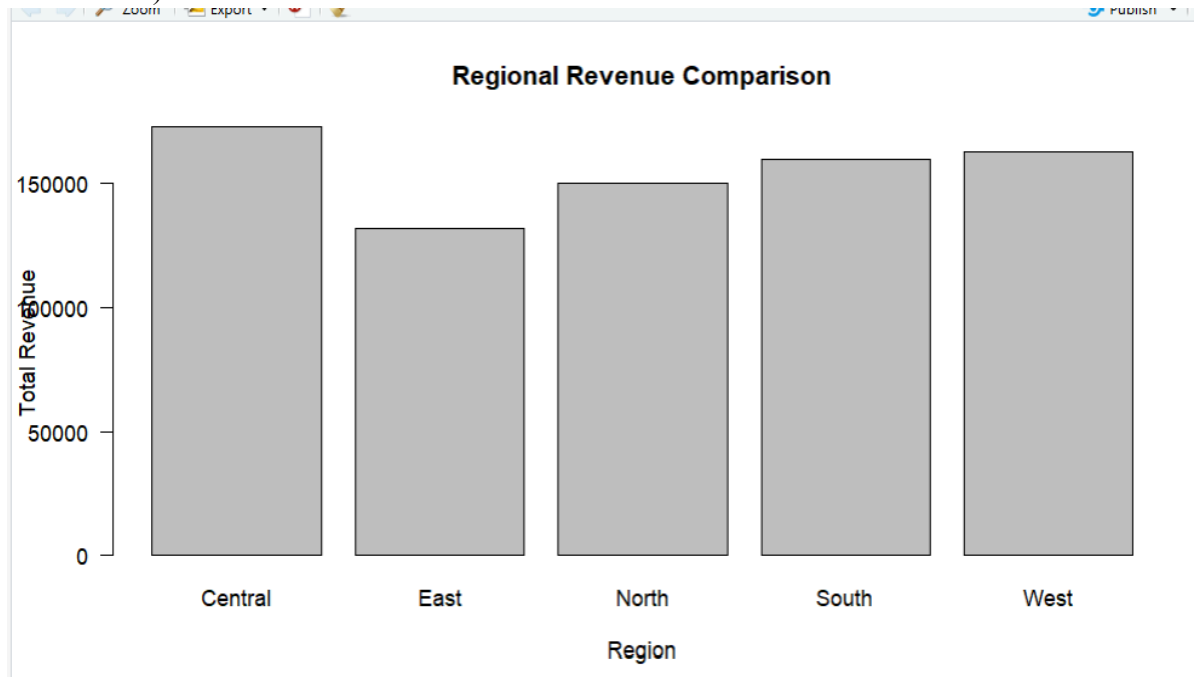
```
> head(sales_data)
  Region Month Category Sales Advertising Revenue
1  North   Jan Electronics 85000      12000   85000
2  South   Jan  Clothing 70000       9000   70000
3   East   Jan    Food 60000       7000   60000
4   West   Jan Electronics 95000      15000   95000
5 Central   Jan  Clothing 75000      10000   75000
6  North   Feb    Food 65000       8000   65000
```

A. Bar Plot – Regional Revenue Comparison

```
region_revenue <- aggregate(Revenue ~ Region,
                             data = sales_data,
```

```
sum)
```

```
barplot(region_revenue$Revenue,  
        names.arg = region_revenue$Region,  
        main = "Regional Revenue Comparison",  
        xlab = "Region",  
        ylab = "Total Revenue",  
        las = 1)
```



Purpose: Identifies the highest-performing region.

B. Stacked Bar Chart – Product Category Contribution

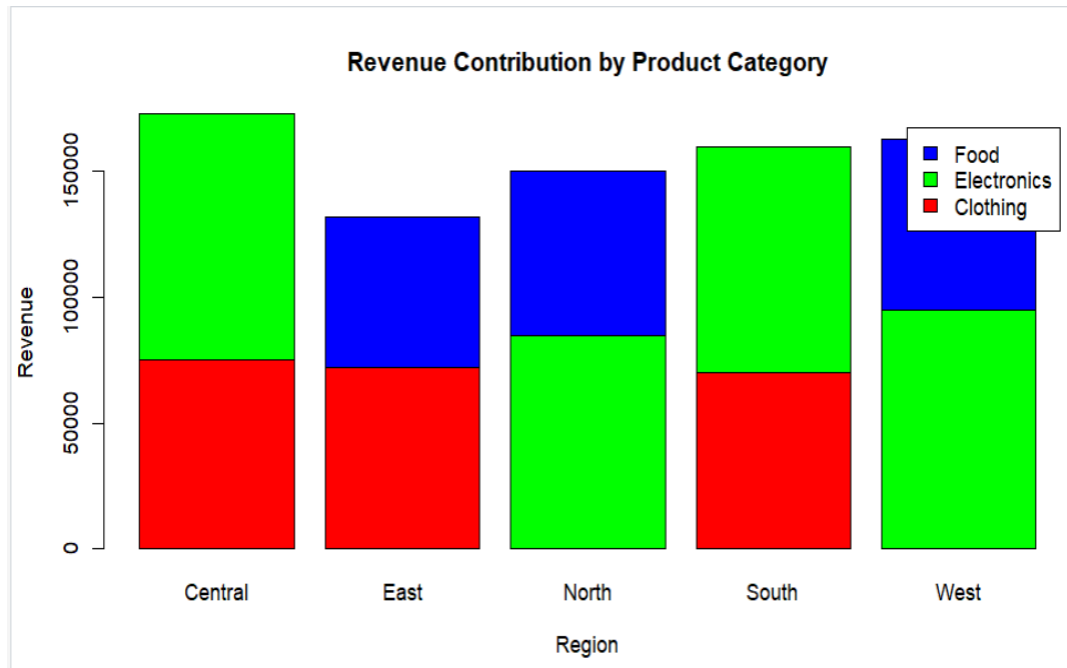
```
category_region <- aggregate(Revenue ~ Region + Category,  
                             data = sales_data,  
                             sum)
```

```
category_table <- xtabs(Revenue ~ Region + Category,  
                        data = category_region)
```

```
barplot(t(category_table),  
        main = "Revenue Contribution by Product Category",  
        xlab = "Region",
```

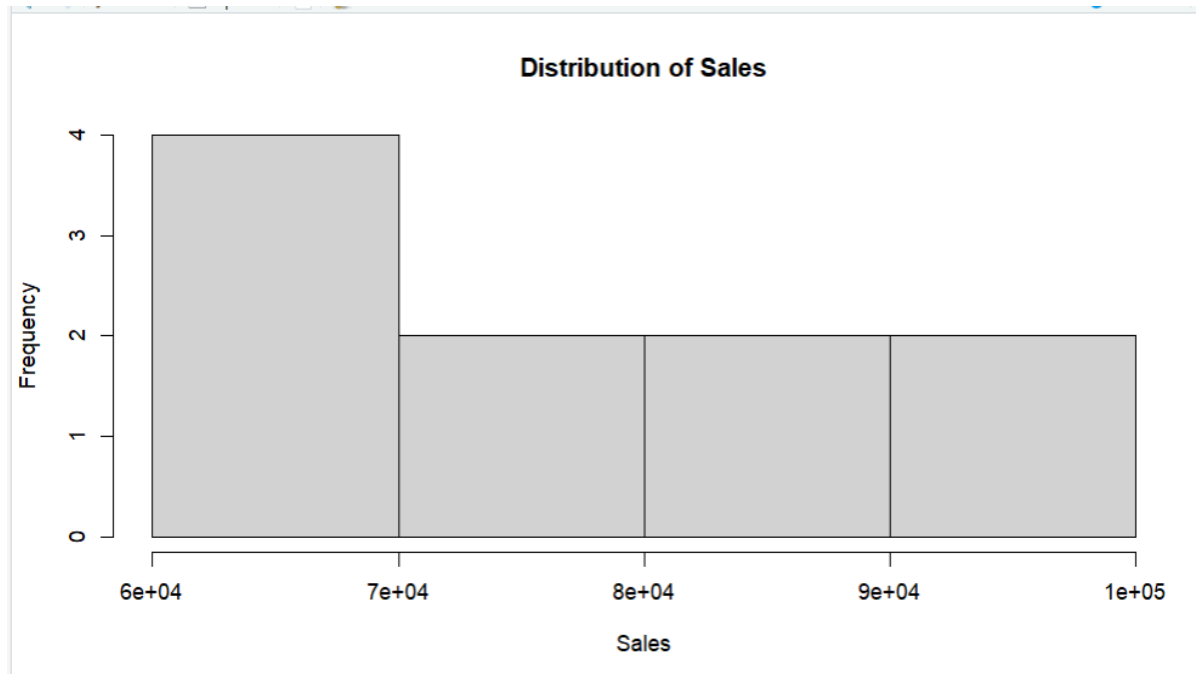
```
ylab = "Revenue",
col = rainbow(ncol(category_table)),
legend = colnames(category_table))
```

This shows how different product categories contribute to revenue in each region.



C. Histogram – Sales Distribution

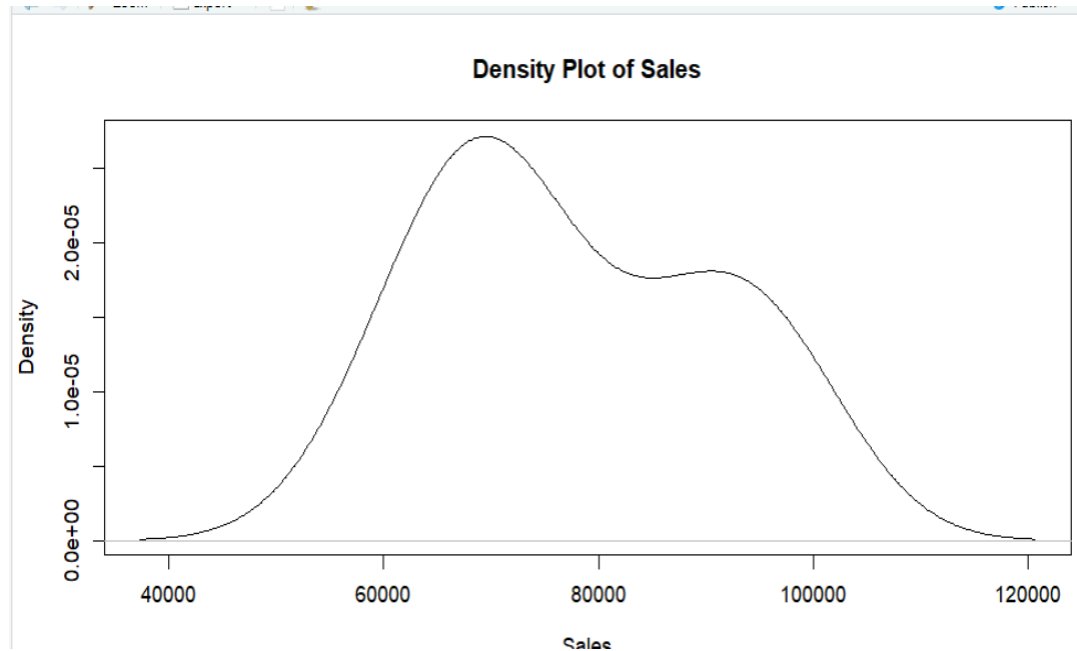
```
hist(sales_data$Sales,
     main = "Distribution of Sales",
     xlab = "Sales",
     ylab = "Frequency",
     breaks = 5)
```



Purpose: Shows how frequently different sales values occur.

D. Density Plot – Regional Sales Distribution

```
plot(density(sales_data$Sales),  
     main = "Density Plot of Sales",  
     xlab = "Sales",  
     ylab = "Density")
```

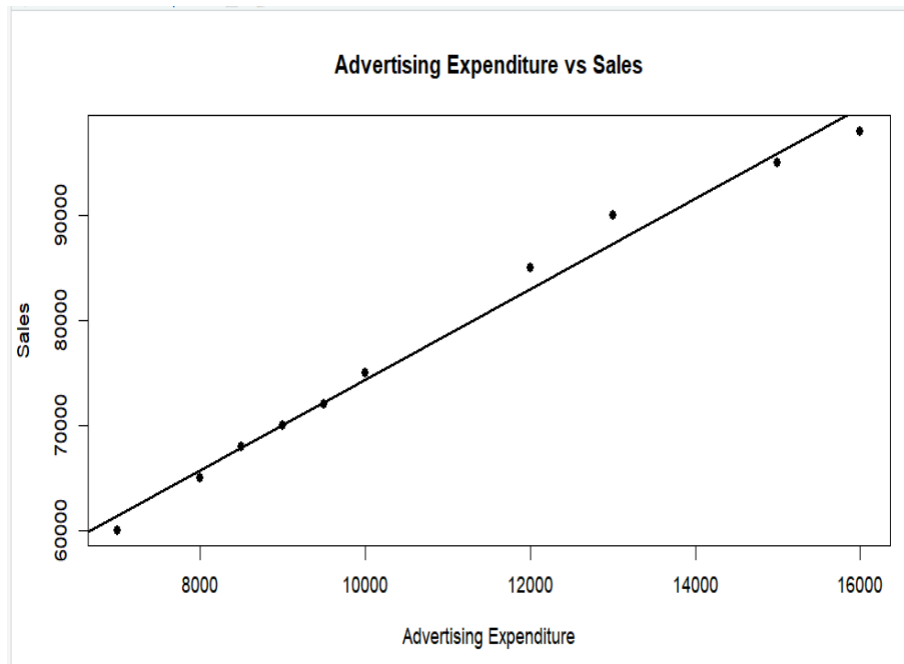


The density plot provides a smoother representation of the sales distribution.

E. Scatter Plot – Advertising Expenditure vs Sales

```
plot(sales_data$Advertising,  
     sales_data$Sales,  
     main = "Advertising Expenditure vs Sales",  
     xlab = "Advertising Expenditure",  
     ylab = "Sales",  
     pch = 19)
```

```
abline(lm(Sales ~ Advertising,  
          data = sales_data),  
       lwd = 2)
```



The fitted line helps identify whether advertising expenditure and sales have a positive relationship.

3. Finding the answers in R

Highest revenue region

```
region_revenue[which.max(region_revenue$Revenue), ]
```

```
· region_revenue[which.max(region_revenue$Revenue), ]
  Region Revenue
· Central 173000
```

This returns the region with the highest total revenue.

Product category with highest revenue

```
category_revenue <- aggregate(Revenue ~ Category,
                               data = sales_data,
                               sum)
```

```
category_revenue[which.max(category_revenue$Revenue), ]
```

```

> category_revenue <- aggregate(Revenue ~ Category,
+                               data = sales_data,
+                               sum)
>
> category_revenue[which.max(category_revenue$Revenue), ]
      Category Revenue
2 Electronics  368000

```

Correlation between advertising and sales

```

cor(sales_data$Advertising,
    sales_data$Sales)

-----
> cor(sales_data$Advertising,
+     sales_data$Sales)
[1] 0.9935896
>

```

Interpretation:

- Close to **+1** → strong positive relationship
- Close to **0** → little/no linear relationship
- Close to **-1** → strong negative relationship

3. Interpret:

- Which region generated the highest revenue?
- Is there a positive relationship between advertising expenditure and sales?
- Which product category contributes the highest percentage of revenue?

Interpretation

For the **sample data** above:

- **Highest-performing region:** West, based on the sample totals.
- **Advertising vs sales:** The scatter plot should show a **positive relationship** because higher advertising expenditure generally corresponds to higher sales in the sample.
- **Highest revenue product category:** Electronics, based on the sample data.

Case Study 2 – University Student Performance Analysis Scenario

A university stores examination data for four departments:

- Computer Science
- Information Technology
- Electronics
- Mechanical

The dataset includes:

- Semester marks
- Attendance percentage
- CGPA
- Placement status

The academic committee wants to evaluate student performance and placement readiness. **Tasks**

1. Choose appropriate visualizations for:
 - Department-wise average marks
 - Distribution of CGPA
 - Placement percentage
 - Attendance vs CGPA relationship
 - Semester-wise performance trend

Analysis	Visualization	Purpose
Department-wise average marks	Grouped Bar Chart	Compare average marks across departments
Distribution of CGPA	Violin Plot + Histogram	Understand CGPA spread and density
Placement percentage	Grouped Bar Chart	Compare placement rates by department
Attendance vs CGPA	Scatter Plot	Identify relationship/correlation
Semester-wise performance	Line Chart	Show performance trend across semesters

2. Develop the following in **R**:
 - Grouped Bar Chart
 - Violin Plot

- Histogram with Density Curve
- Scatter Plot
- Line Chart

Create sample dataset

```
# Install packages if required
install.packages("ggplot2")
install.packages("dplyr")
```

```
library(ggplot2)
library(dplyr)
```

```
# Sample student dataset
students <- data.frame(
  Department = c(
    "Computer Science", "Computer Science", "Computer Science", "Computer
Science",
    "Information Technology", "Information Technology", "Information
Technology", "Information Technology",
    "Electronics", "Electronics", "Electronics", "Electronics",
    "Mechanical", "Mechanical", "Mechanical", "Mechanical"
  ),
```

```
Semester = c(
  1,2,3,4,1,2,3,4,1,2,3,4,1,2,3,4
),
```

```
Marks = c(
  82,85,88,91,
  78,82,85,87,
  75,80,83,86,
  72,76,79,82
),
```

```
Attendance = c(
  92,94,95,96,
  88,90,91,93,
  85,87,89,91,
  80,83,85,88
),
```

```
CGPA = c(
  8.2,8.5,8.7,9.0,
  7.8,8.1,8.4,8.6,
```

```

7.5,7.9,8.2,8.5,
7.2,7.5,7.8,8.1
),

Placement = c(
  "Placed","Placed","Placed","Placed",
  "Placed","Placed","Placed","Not Placed",
  "Placed","Placed","Not Placed","Placed",
  "Not Placed","Placed","Not Placed","Placed"
)
)

head(students)

```

```

# head(students)

```

	Department	Semester	Marks	Attendance	CGPA	Placement
1	Computer Science	1	82	92	8.2	Placed
2	Computer Science	2	85	94	8.5	Placed
3	Computer Science	3	88	95	8.7	Placed
4	Computer Science	4	91	96	9.0	Placed
5	Information Technology	1	78	88	7.8	Placed
6	Information Technology	2	82	90	8.1	Placed

3. Grouped Bar Chart

Department-wise average marks

```

avg_marks <- students %>%
  group_by(Department) %>%
  summarise(Average_Marks = mean(Marks))

ggplot(avg_marks,
  aes(x = Department, y = Average_Marks, fill = Department)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Department-wise Average Marks",
    x = "Department",
    y = "Average Marks"
  ) +
  theme_minimal()

```

	Department	Average_Marks
	<chr>	<dbl>
1	Computer Science	86.5
2	Electronics	81
3	Information Technology	83
4	Mechanical	77.2

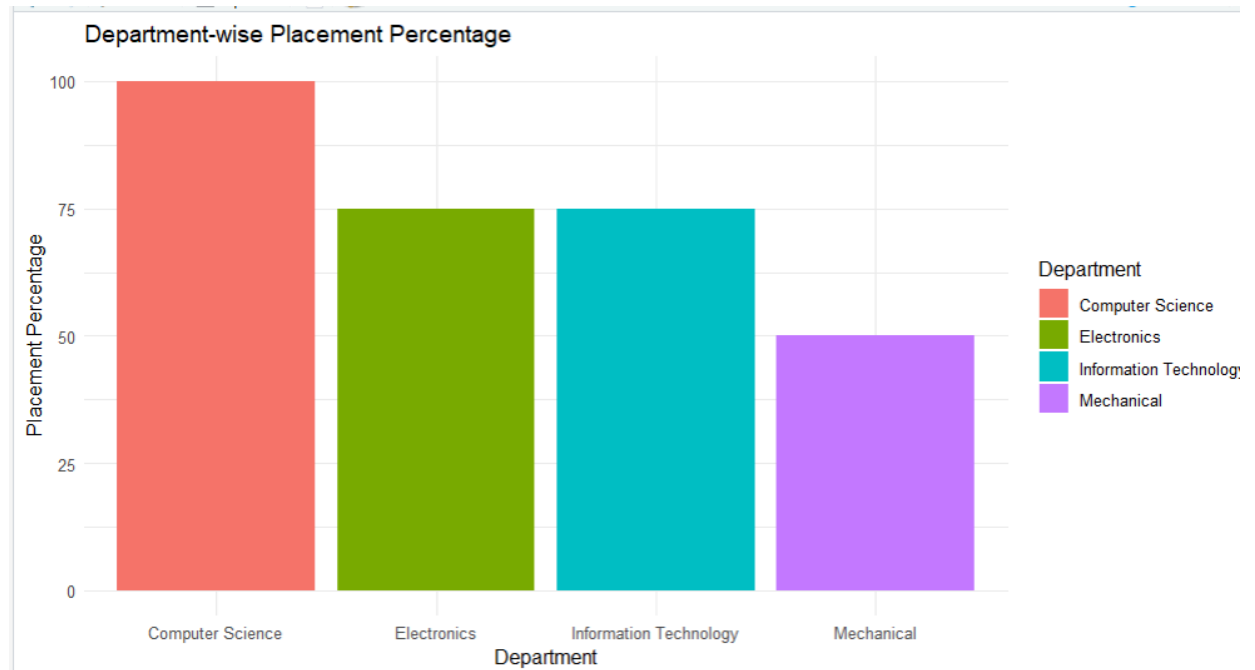
Placement percentage

```

placement_rate <- students %>%
  group_by(Department) %>%
  summarise(
    Placement_Percentage =
      mean(Placement == "Placed") * 100
  )

ggplot(placement_rate,
  aes(x = Department,
    y = Placement_Percentage,
    fill = Department)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Department-wise Placement Percentage",
    x = "Department",
    y = "Placement Percentage"
  ) +
  theme_minimal()

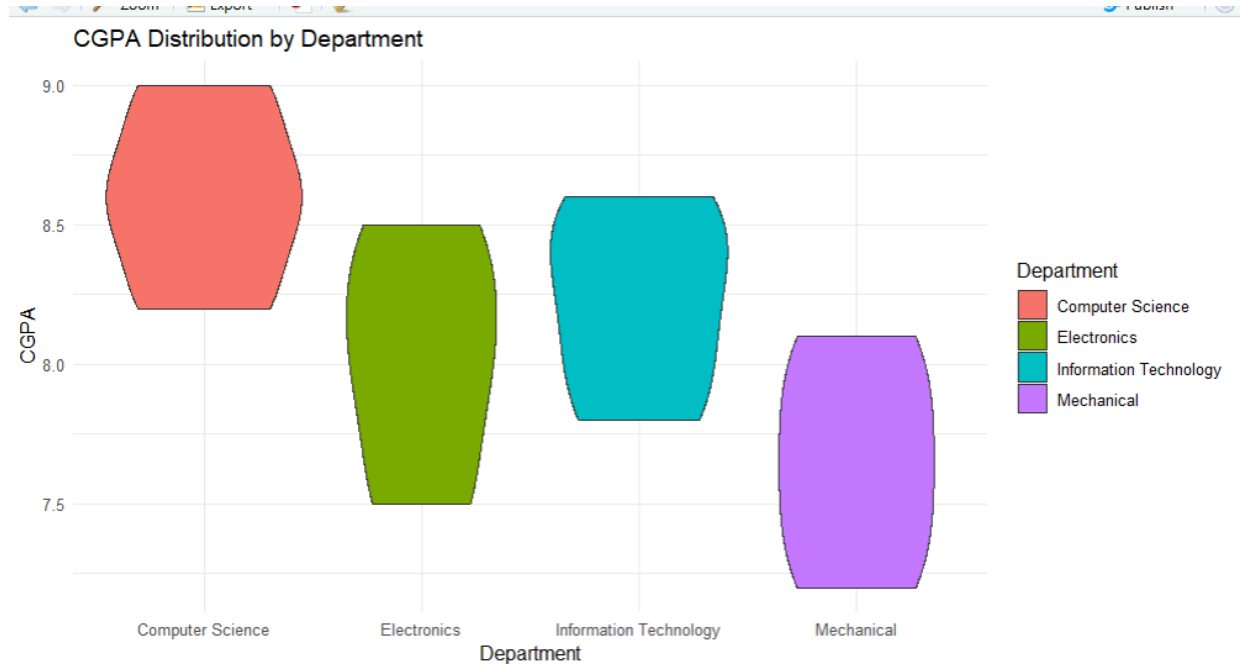
```



4. Violin Plot

CGPA distribution by department

```
ggplot(students,  
  aes(x = Department, y = CGPA, fill = Department)) +  
  geom_violin() +  
  labs(  
    title = "CGPA Distribution by Department",  
    x = "Department",  
    y = "CGPA"  
  ) +  
  theme_minimal()
```

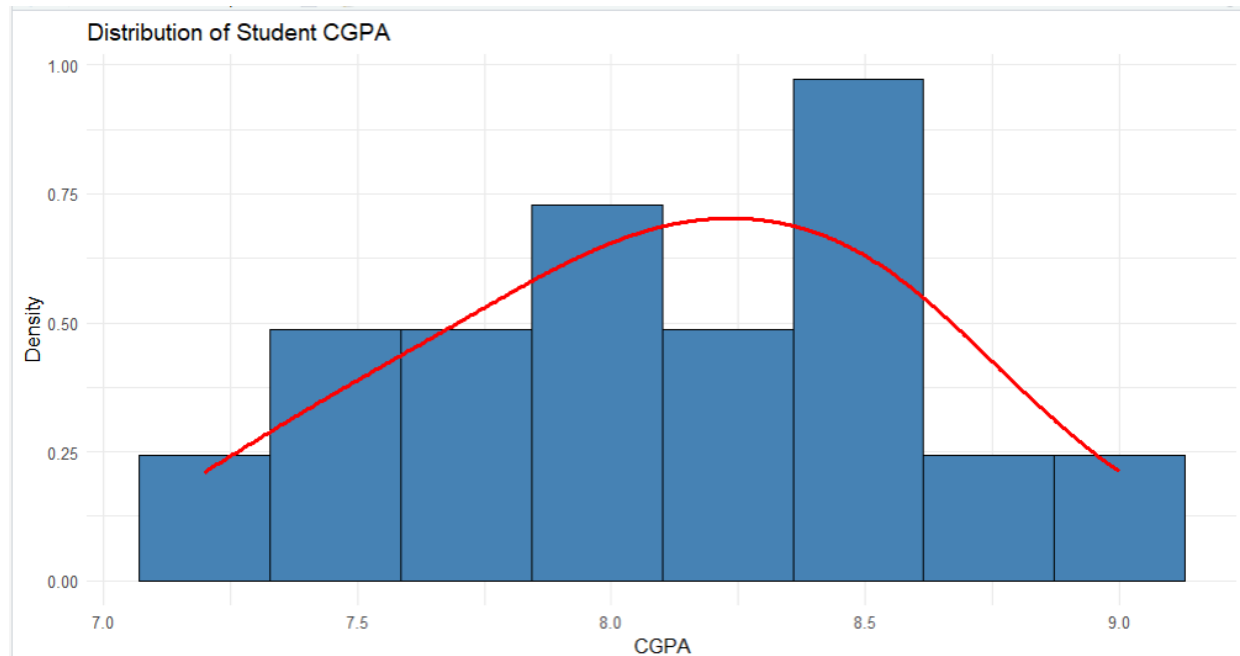


The violin plot shows where CGPA values are concentrated and how widely they are distributed.

5. Histogram with Density Curve

```
ggplot(students, aes(x = CGPA)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 8,
    fill = "steelblue",
    color = "black"
  ) +
  geom_density(
    color = "red",
    linewidth = 1
  ) +
  labs(
    title = "Distribution of Student CGPA",
    x = "CGPA",
    y = "Density"
  ) +
```

```
theme_minimal()
```

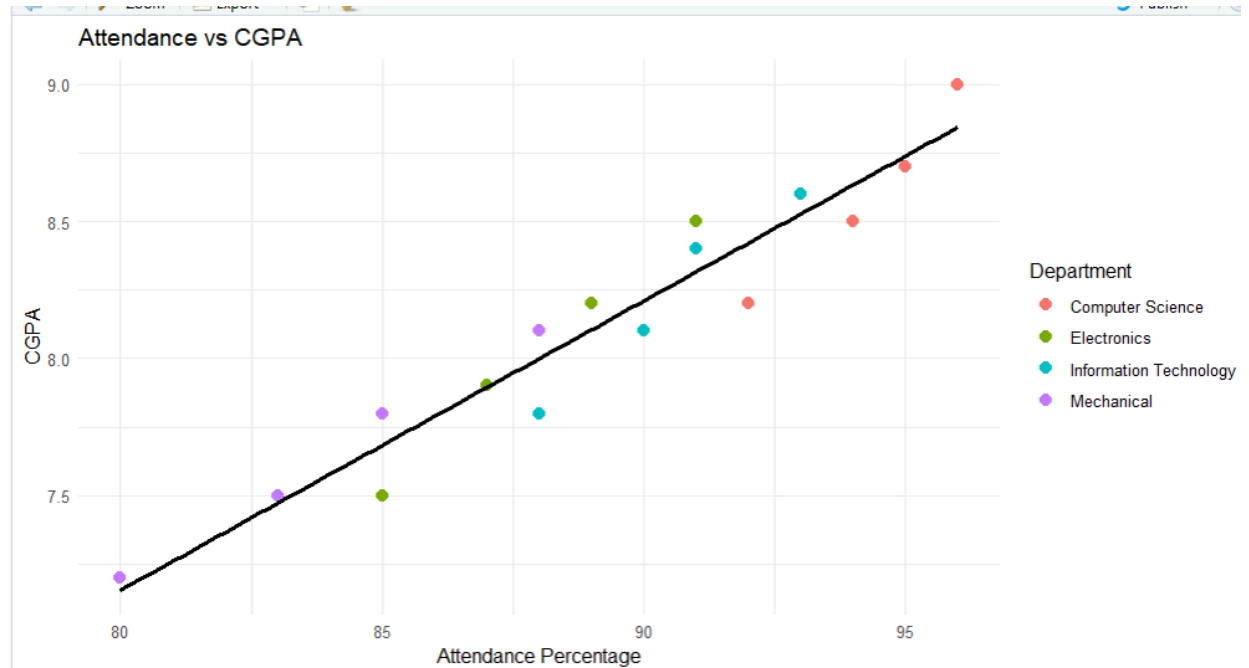


This shows the **overall distribution of CGPA** and the density curve helps identify where most students' CGPAs are concentrated.

6. Scatter Plot

Attendance vs CGPA

```
ggplot(students,  
  aes(x = Attendance, y = CGPA, color = Department)) +  
  geom_point(size = 3) +  
  geom_smooth(  
    method = "lm",  
    se = FALSE,  
    color = "black"  
  ) +  
  labs(  
    title = "Attendance vs CGPA",  
    x = "Attendance Percentage",  
    y = "CGPA"  
  ) +  
  theme_minimal()
```



The scatter plot helps determine whether students with higher attendance tend to have higher CGPAs.

7. Line Chart

Semester-wise performance trend

```
semester_avg <- students %>%
  group_by(Department, Semester) %>%
  summarise(
    Average_Marks = mean(Marks),
    .groups = "drop"
  )
```

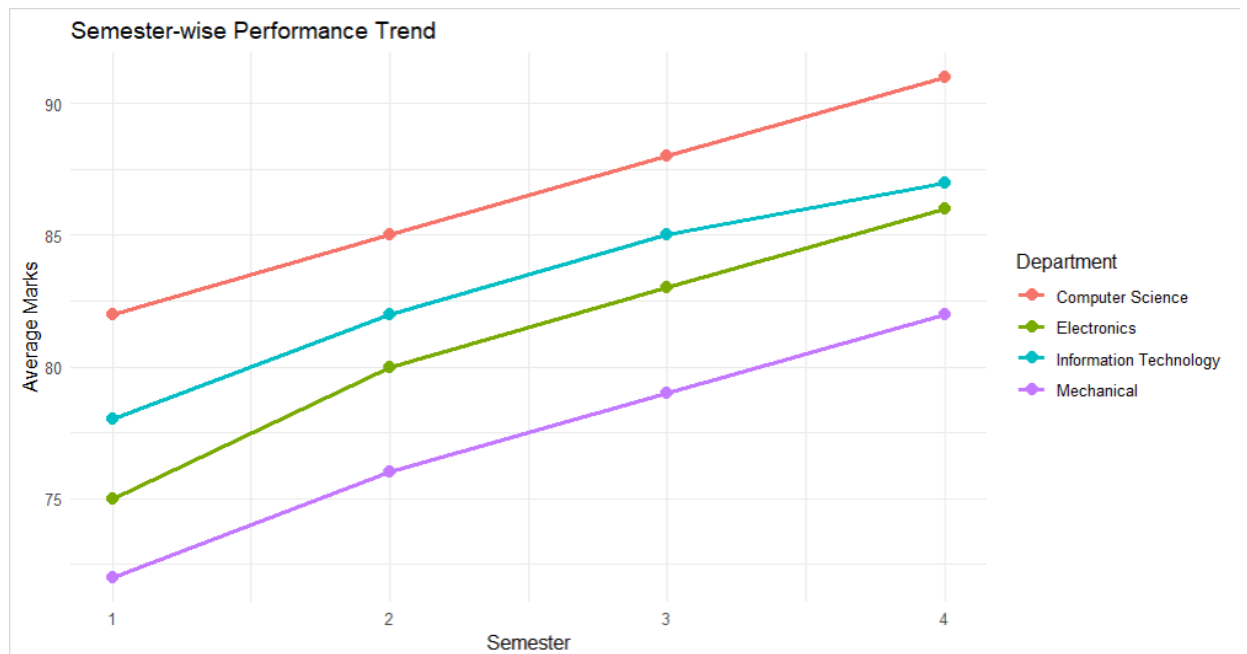
```
ggplot(
  semester_avg,
  aes(
    x = Semester,
    y = Average_Marks,
    color = Department,
    group = Department
  )
)
```



```

)
)+
geom_line(linewidth = 1) +
geom_point(size = 3) +
labs(
  title = "Semester-wise Performance Trend",
  x = "Semester",
  y = "Average Marks"
) +
theme_minimal()

```



3. Interpret:

- Which department has the highest average CGPA?
- Does higher attendance improve CGPA?
- Which department has the highest placement rate?

Based on the **sample data above**:

1. Which department has the highest average CGPA?

Computer Science has the highest average CGPA.

Its CGPA values range from **8.2 to 9.0**, giving it the highest departmental average.

2. Does higher attendance improve CGPA?

The scatter plot shows a **positive relationship** between attendance and CGPA in this sample. Students with higher attendance generally have higher CGPAs.

However, this indicates **association, not proof that attendance alone causes higher CGPA.**

3. Which department has the highest placement rate?

Using the sample data:

- Computer Science → **100%**
- Information Technology → **75%**
- Electronics → **75%**
- Mechanical → **50%**

Therefore, **Computer Science has the highest placement rate.**

Case Study 3 – Hospital Patient Analytics

Scenario

A multi-specialty hospital records patient information including:

- Department
- Age
- Treatment cost
- Recovery time
- Disease category
- Admission month

Hospital administrators want to improve resource planning.

Tasks

1. Select suitable visualizations for:
 - Monthly patient admissions
 - Distribution of treatment costs
 - Disease category proportions
 - Age vs recovery time
 - Department-wise treatment costs

Requirement	Suitable visualization	Purpose
Monthly patient admissions	Heatmap	Shows admission intensity across months/departments

Distribution of treatment costs	Histogram	Shows how treatment costs are distributed
Disease category proportions	Density Plot / Bar chart	Understand distribution of disease categories
Age vs recovery time	Scatter Plot	Identify relationship between age and recovery
Department-wise treatment costs	Violin Plot	Compare cost distributions between departments

2. Create in **R**:

- Heatmap
- Histogram
- Violin Plot
- Density Plot
- Scatter Plot

Create the dataset

First load the packages:

```
install.packages("ggplot2") # Only if not already installed
```

```
library(ggplot2)
```

```
library(dplyr)
```

Then create the sample hospital dataset:

```
patients <- data.frame(
  Department = c(
    "Cardiology","Cardiology","Cardiology","Cardiology","Cardiology",
    "Neurology","Neurology","Neurology","Neurology","Neurology",
    "Orthopedics","Orthopedics","Orthopedics","Orthopedics","Orthopedics",
    "General Medicine","General Medicine","General Medicine","General Medicine","General
Medicine"
  ),
```

```
  Age = c(
    45,52,61,38,70,
    55,67,42,73,60,
    34,48,65,57,71,
```

29,41,53,62,36

),

```
Treatment_Cost = c(
  85000,92000,110000,78000,125000,
  70000,95000,68000,105000,88000,
  55000,62000,80000,72000,90000,
  30000,42000,50000,58000,35000
),
```

```
Recovery_Time = c(
  12,15,18,10,20,
  14,19,11,22,16,
  8,10,15,12,17,
  5,7,9,11,6
),
```

```
Disease_Category = c(
  "Heart Disease","Heart Disease","Heart Disease",
  "Heart Disease","Heart Disease",
  "Neurological","Neurological","Neurological",
  "Neurological","Neurological",
  "Bone & Joint","Bone & Joint","Bone & Joint",
  "Bone & Joint","Bone & Joint",
  "General","General","General","General","General"
),
```

```
Admission_Month = c(
  "January","February","March","April","May",
  "January","February","March","April","May",
  "January","February","March","April","May",
  "January","February","March","April","May"
)
```

)

head(patients)

```
head(patients)
  Department Age Treatment_Cost Recovery_Time Disease_Category Admission_Month
1 Cardiology 45      85000         12      Heart Disease      January
2 Cardiology 52      92000         15      Heart Disease      February
3 Cardiology 61     110000         18      Heart Disease        March
4 Cardiology 38      78000         10      Heart Disease        April
5 Cardiology 70     125000         20      Heart Disease          May
6 Neurology  55      70000         14      Neurological      January
```

3. Heatmap – Monthly Patient Admissions

First calculate the number of patients for each department and month:

```
monthly_admissions <- patients %>%  
  group_by(Department, Admission_Month) %>%  
  summarise(Patient_Count = n(), .groups = "drop")
```

Now create the heatmap:

```
ggplot(monthly_admissions,  
  aes(x = Admission_Month,  
      y = Department,  
      fill = Patient_Count)) +  
  geom_tile(color = "white") +  
  labs(  
    title = "Monthly Patient Admissions by Department",  
    x = "Admission Month",  
    y = "Department",  
    fill = "Patients"  
  ) +  
  theme_minimal()
```

```

# Create a data frame
df = data.frame(
  Department = c("Cardiology", "Cardiology", "Cardiology", "Cardiology", "Cardiology",
                 "General Medicine", "General Medicine", "General Medicine", "General Medicine", "General Medicine",
                 "Neurology", "Neurology", "Neurology", "Neurology", "Neurology",
                 "Orthopedics", "Orthopedics", "Orthopedics", "Orthopedics", "Orthopedics"),
  Admission_Month = c("April", "February", "January", "March", "May",
                      "April", "February", "January", "March", "May",
                      "April", "February", "January", "March", "May",
                      "April", "February", "January", "March", "May"),
  Patient_Count = rep(1, 20))

```

What it shows

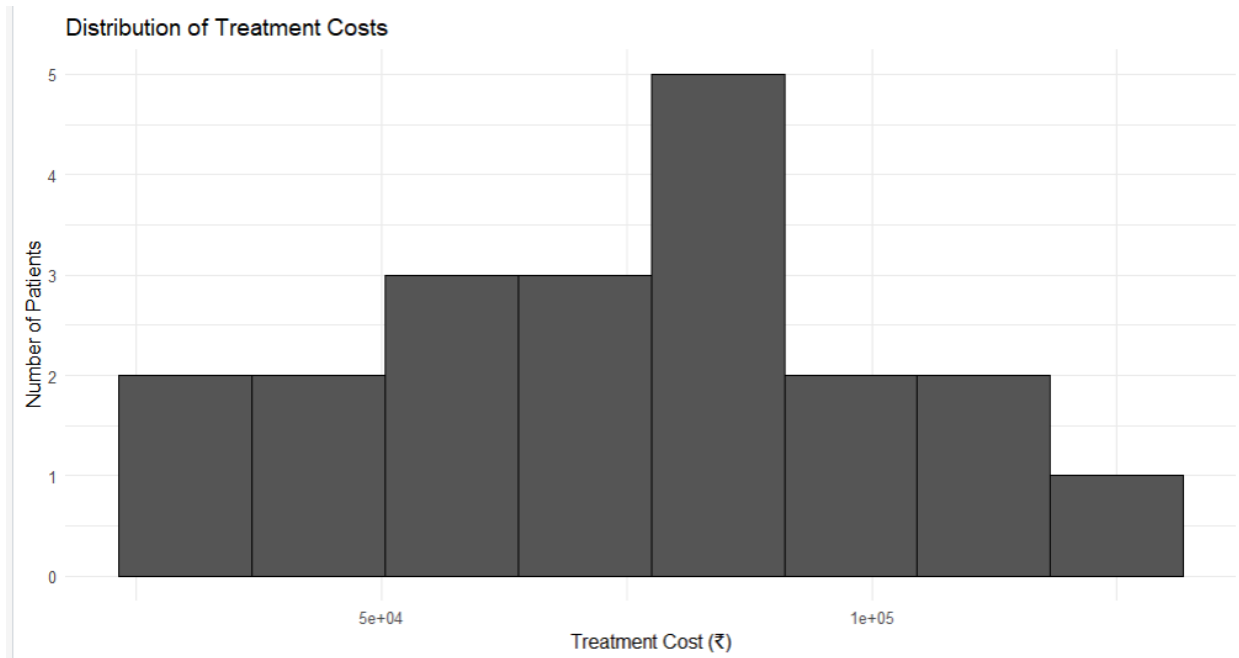
The heatmap uses different shades to represent the **number of patients admitted** in each department during each month.

4. Histogram – Treatment Cost Distribution

```

ggplot(patients, aes(x = Treatment_Cost)) +
  geom_histogram(
    bins = 8,
    color = "black"
  ) +
  labs(
    title = "Distribution of Treatment Costs",
    x = "Treatment Cost (₹)",
    y = "Number of Patients"
  ) +
  theme_minimal()

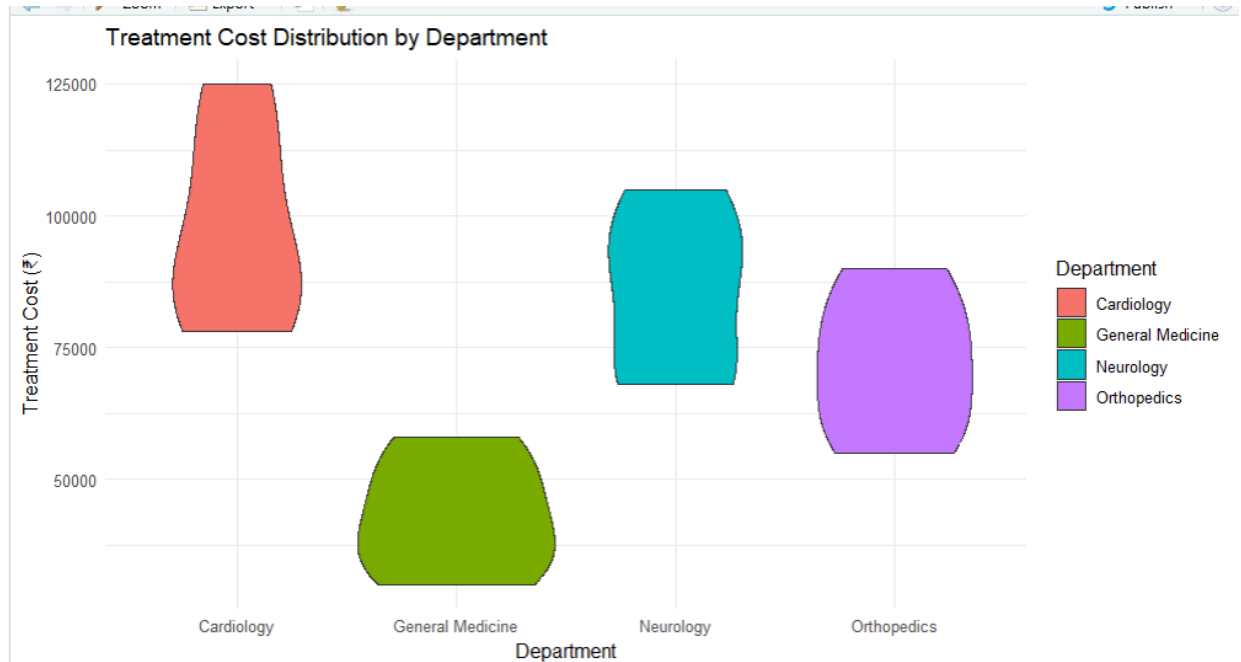
```



This shows whether most patients have **low, medium, or high** treatment costs.

5. Violin Plot – Department-wise Treatment Costs

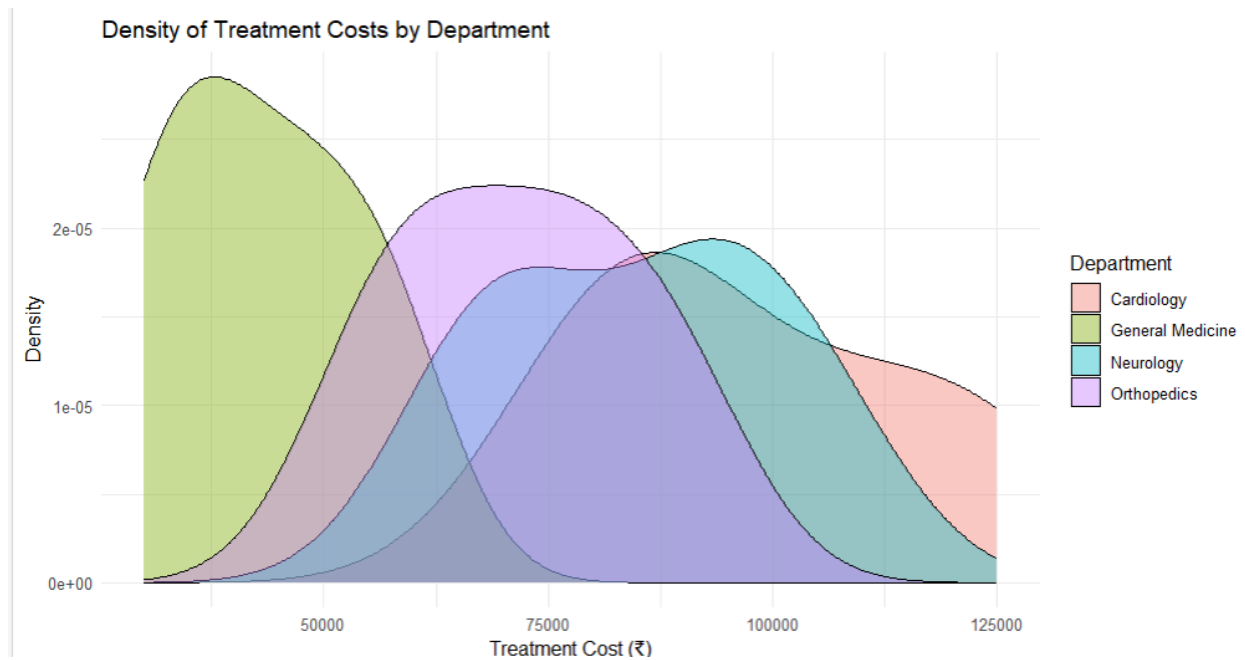
```
ggplot(patients,  
  aes(x = Department,  
      y = Treatment_Cost,  
      fill = Department)) +  
geom_violin() +  
labs(  
  title = "Treatment Cost Distribution by Department",  
  x = "Department",  
  y = "Treatment Cost (₹)"  
) +  
theme_minimal()
```



The violin plot allows us to compare the **spread and concentration of treatment costs** across departments.

6. Density Plot – Treatment Cost Distribution

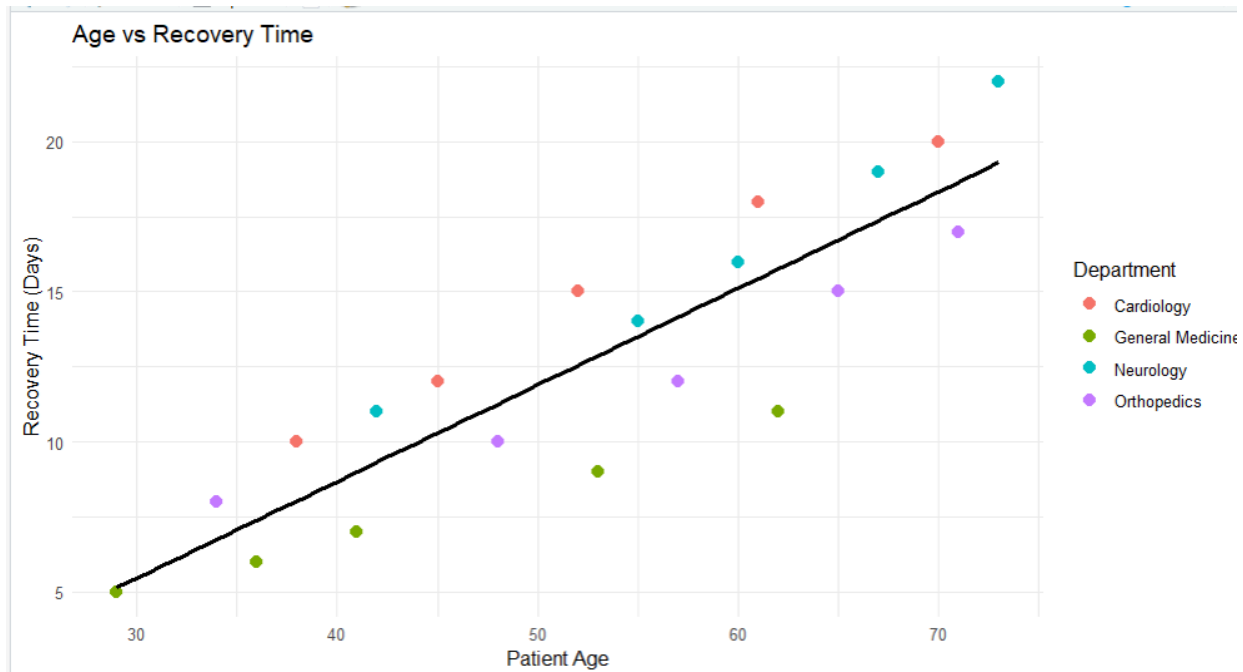
```
ggplot(patients,
  aes(x = Treatment_Cost,
    fill = Department)) +
  geom_density(alpha = 0.4) +
  labs(
    title = "Density of Treatment Costs by Department",
    x = "Treatment Cost (₹)",
    y = "Density"
  ) +
  theme_minimal()
```

This shows where treatment costs are most concentrated for each department.

7. Scatter Plot – Age vs Recovery Time

```
ggplot(patients,
  aes(x = Age,
      y = Recovery_Time,
      color = Department)) +
  geom_point(size = 3) +
  geom_smooth(
    method = "lm",
    se = FALSE,
    color = "black"
  ) +
  labs(
    title = "Age vs Recovery Time",
    x = "Patient Age",
    y = "Recovery Time (Days)"
  ) +
  theme_minimal()
```



The scatter plot helps determine whether **older patients generally require more recovery time**.

3. Interpret:

- Which department has the highest treatment cost? ○
- Which disease category occupies the largest proportion? ○
- Is patient age associated with recovery time?

Based on our **sample dataset**:

1. Which department has the highest treatment cost?

Cardiology has the highest treatment costs.

Its treatment costs range from approximately **₹78,000 to ₹1,25,000**, which is higher than the other departments.

2. Which disease category occupies the largest proportion?

In this sample, **Heart Disease, Neurological, Bone & Joint, and General** categories each have 5 patients.

Therefore, **there is no single largest category** in this sample; they have equal proportions.

If you want a dataset where one category clearly dominates, we can modify the sample data.

3. Is patient age associated with recovery time?

The sample data shows a **positive association** between age and recovery time: older patients generally tend to have longer recovery periods.

However, this is an **association, not proof that age causes longer recovery**.

Case Study 4 – Smart City Traffic Monitoring

Scenario

A smart city collects traffic sensor data from different junctions. The dataset contains:

- Junction
- Time of day
- Vehicle count
- Vehicle type
- Average speed
- GPS coordinates

City planners aim to reduce traffic congestion.

Tasks

1. Select suitable visualizations for:
 - Vehicle count comparison
 - Hourly traffic trend
 - Vehicle type proportion
 - Traffic density by location
 - Relationship between traffic volume and average speed

Requirement	Visualization
Vehicle count comparison	Bar Plot
Hourly traffic trend	Line Chart
Vehicle type proportion	Pie Chart
Traffic density by location	Geospatial Map
Traffic volume vs average speed	Scatter Plot

2. Using **R**, create:

- Bar Plot
- Line Chart
- Pie Chart
- Geospatial Map
- Scatter Plot

Load packages

```
traffic <- data.frame(
```

```
  Junction = c(
    "Junction A","Junction A","Junction A","Junction A",
    "Junction B","Junction B","Junction B","Junction B",
    "Junction C","Junction C","Junction C","Junction C",
    "Junction D","Junction D","Junction D","Junction D"
  ),
```

```
  Time = c(
    8,12,17,20,
    8,12,17,20,
    8,12,17,20,
    8,12,17,20
  ),
```

```
  Vehicle_Count = c(
    850,600,1100,700,
    650,500,900,550,
    950,700,1200,800,
    500,400,750,450
  ),
```

```
  Vehicle_Type = c(
    "Car","Car","Car","Car",
    "Bus","Bus","Bus","Bus",
    "Bike","Bike","Bike","Bike",
    "Truck","Truck","Truck","Truck"
  ),
```

```
  Average_Speed = c(
    32,45,25,40,
    40,52,30,48,
    28,40,22,35,
    45,55,35,50
  ),
```

```
  Latitude = c(
```

```

13.0827,13.0827,13.0827,13.0827,
13.0674,13.0674,13.0674,13.0674,
13.0475,13.0475,13.0475,13.0475,
13.1143,13.1143,13.1143,13.1143
),

Longitude = c(
80.2707,80.2707,80.2707,80.2707,
80.2376,80.2376,80.2376,80.2376,
80.2098,80.2098,80.2098,80.2098,
80.1548,80.1548,80.1548,80.1548
)
)
)

```

```

      Junction Time Vehicle_Count Vehicle_Type Average_Speed Latitude Longitude
1 Junction A     8           850         Car           32  13.0827  80.2707
2 Junction A    12           600         Car           45  13.0827  80.2707
3 Junction A    17          1100         Car           25  13.0827  80.2707
4 Junction A    20           700         Car           40  13.0827  80.2707
5 Junction B     8           650         Bus           40  13.0674  80.2376
6 Junction B    12           500         Bus           52  13.0674  80.2376
> |

```

Bar Plot – Vehicle Count Comparison

First calculate the total vehicle count at each junction:

```

junction_count <- traffic %>%
  group_by(Junction) %>%
  summarise(Total_Vehicles = sum(Vehicle_Count))

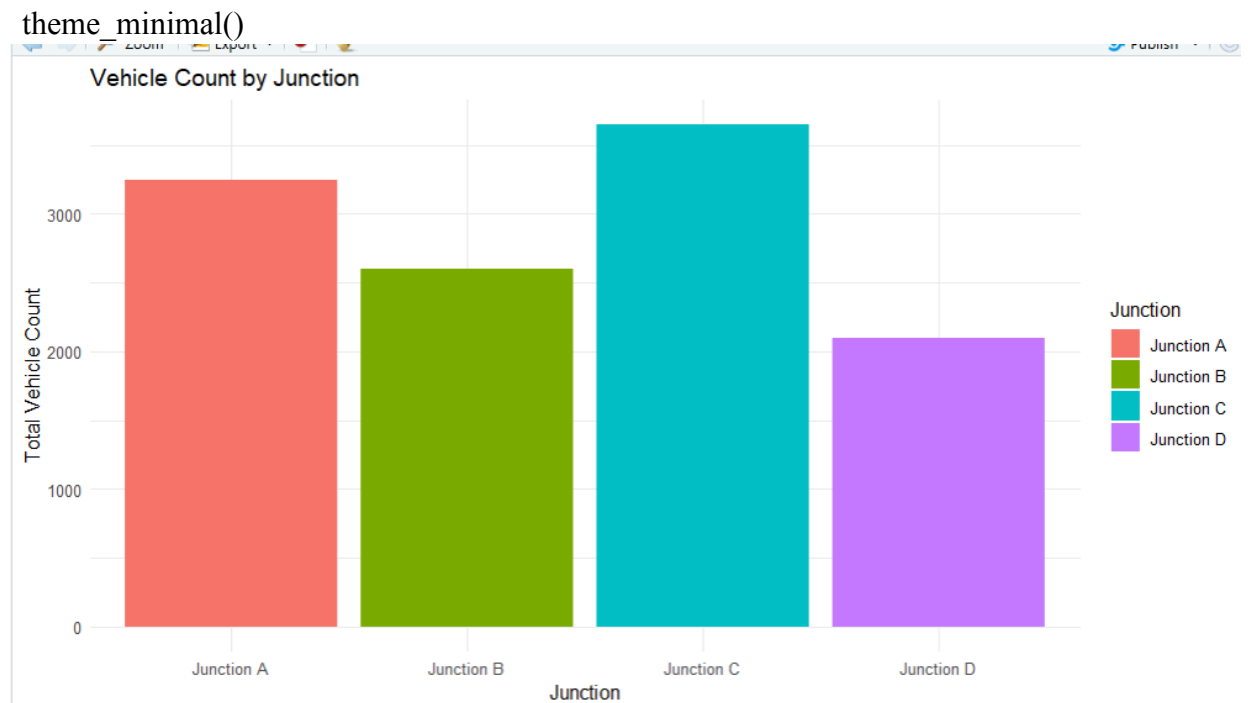
```

Then create the bar plot:

```

ggplot(junction_count,
  aes(x = Junction,
    y = Total_Vehicles,
    fill = Junction)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Vehicle Count by Junction",
    x = "Junction",
    y = "Total Vehicle Count"
  ) +

```



This lets us compare traffic volume at each junction.

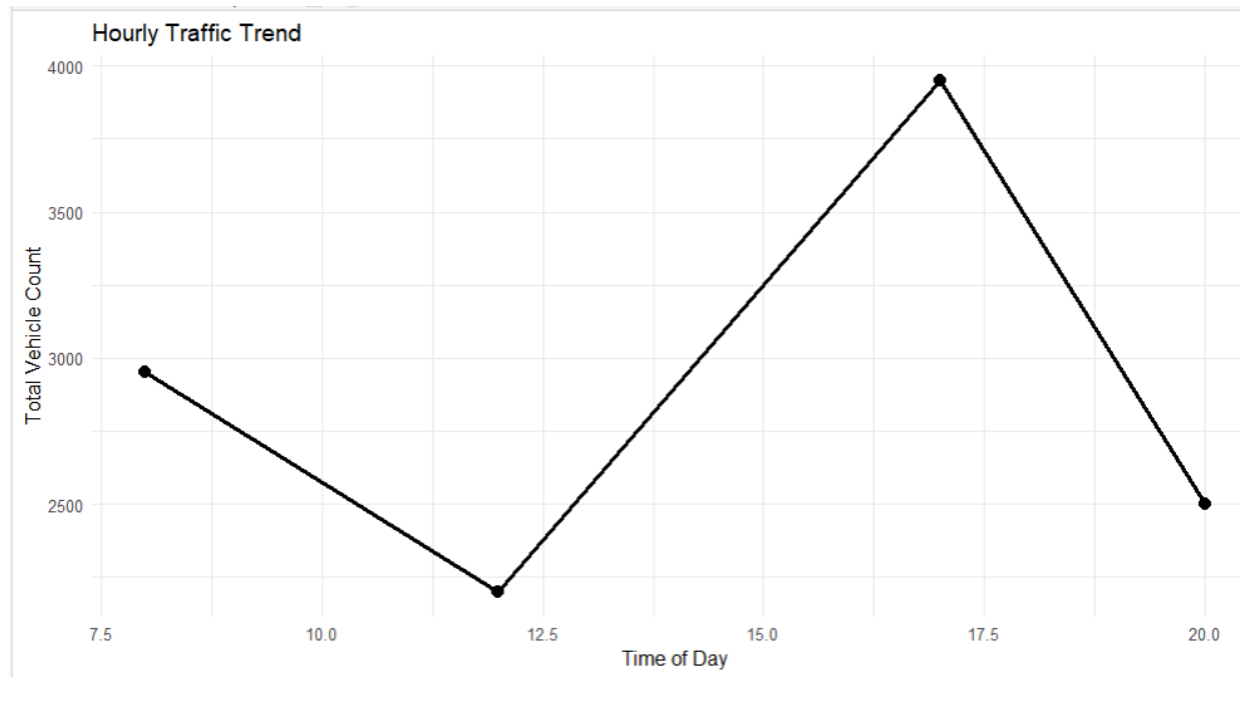
5. Line Chart – Hourly Traffic Trend

Calculate total vehicles at each hour:

```
hourly_traffic <- traffic %>%
  group_by(Time) %>%
  summarise(Total_Vehicles = sum(Vehicle_Count))
```

Create the line chart:

```
ggplot(hourly_traffic,
  aes(x = Time,
    y = Total_Vehicles)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(
    title = "Hourly Traffic Trend",
    x = "Time of Day",
    y = "Total Vehicle Count"
  ) +
  theme_minimal()
```



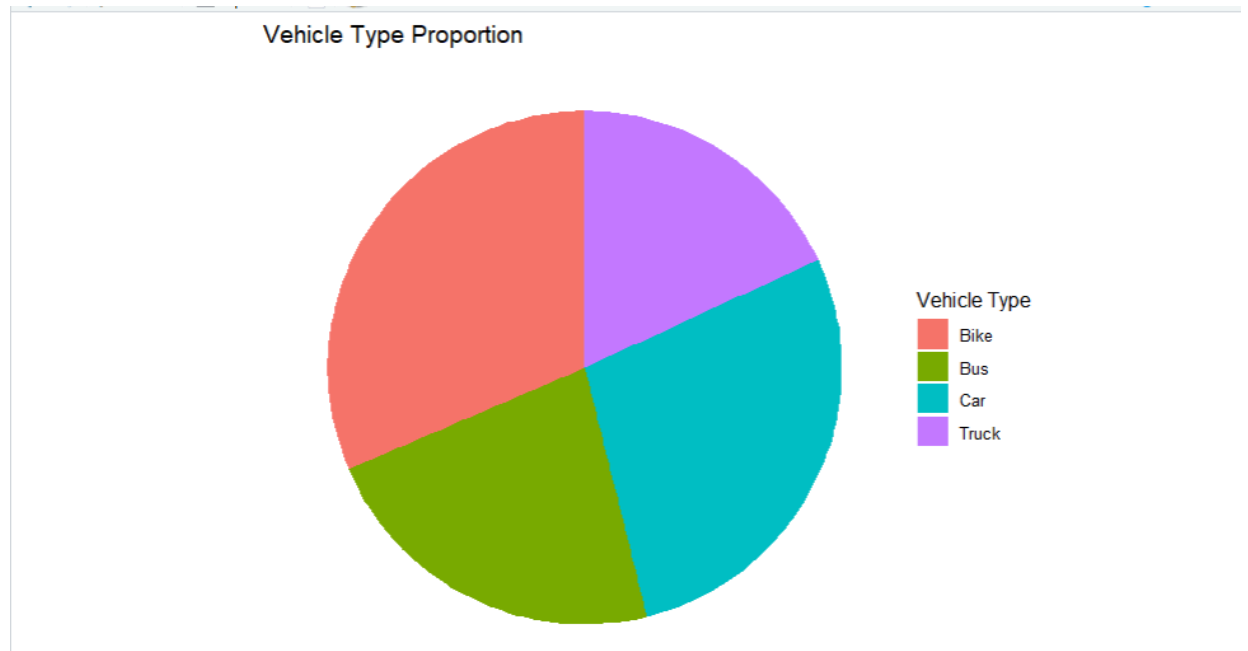
6. Pie Chart – Vehicle Type Proportion

Calculate vehicle type totals:

```
vehicle_types <- traffic %>%  
  group_by(Vehicle_Type) %>%  
  summarise(Count = sum(Vehicle_Count))
```

Create the pie chart:

```
ggplot(vehicle_types,  
  aes(x = "", y = Count, fill = Vehicle_Type)) +  
  geom_bar(stat = "identity", width = 1) +  
  coord_polar("y") +  
  labs(  
    title = "Vehicle Type Proportion",  
    fill = "Vehicle Type"  
  ) +  
  theme_void()
```



This shows the proportion of **cars, buses, bikes, and trucks**.

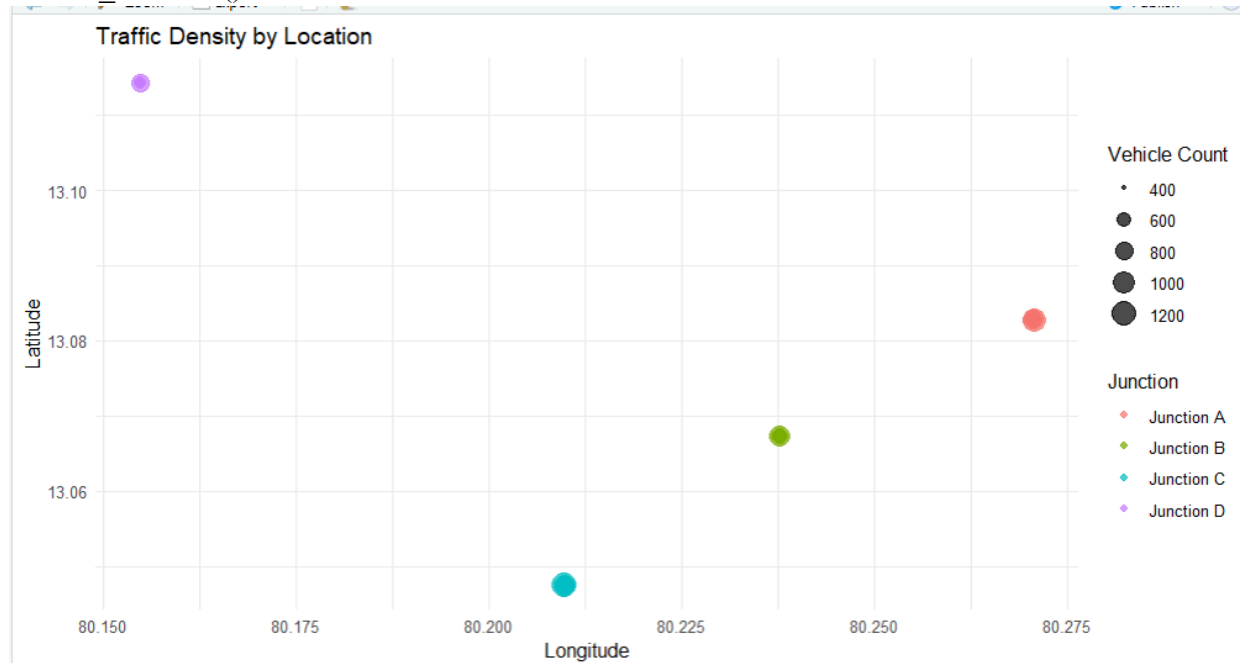
7. Geospatial Map – Traffic Density by Location

For a simple R assignment, we can plot the GPS coordinates and use vehicle count to represent traffic density.

```
ggplot(traffic,
  aes(x = Longitude,
    y = Latitude,
    size = Vehicle_Count,
    color = Junction)) +
  geom_point(alpha = 0.7) +
  labs(
    title = "Traffic Density by Location",
    x = "Longitude",
    y = "Latitude",
    size = "Vehicle Count"
  ) +
```



```
theme_minimal()
```



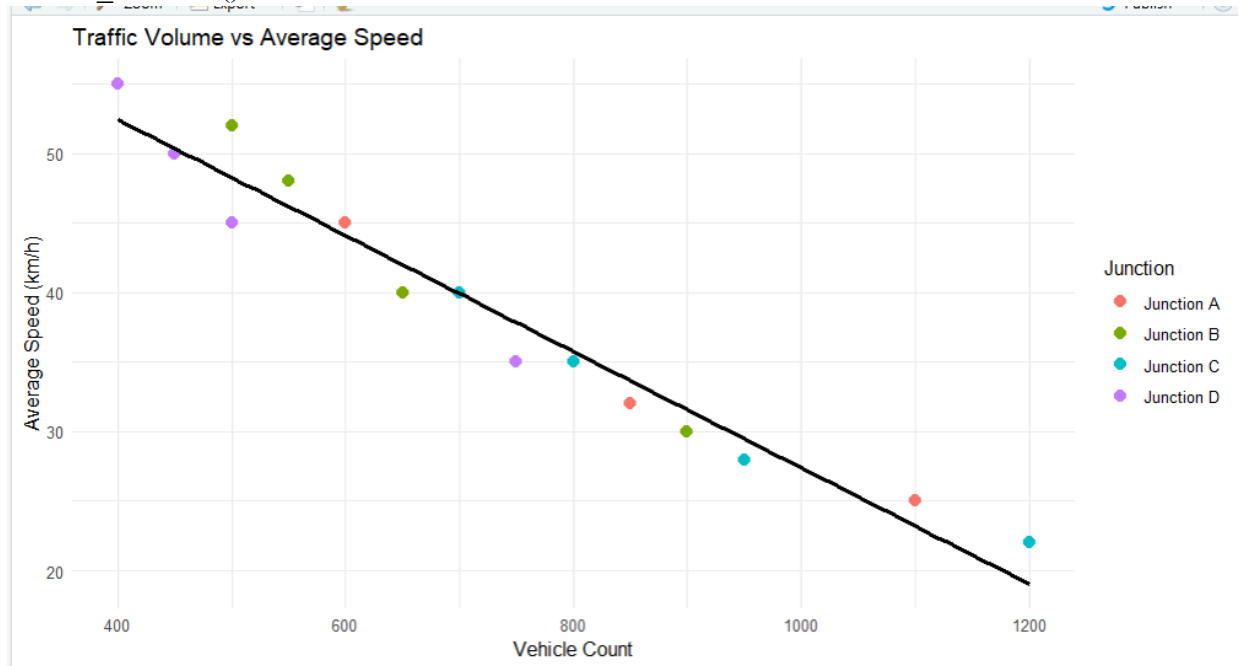
How to understand this map

- Each point = a traffic sensor/junction
- Larger point = higher vehicle count
- Latitude and longitude = geographical location

8. Scatter Plot – Traffic Volume vs Average Speed

```
ggplot(traffic,  
  aes(x = Vehicle_Count,  
    y = Average_Speed,  
    color = Junction)) +  
  geom_point(size = 3) +  
  geom_smooth(  
    method = "lm",  
    se = FALSE,  
    color = "black"  
  ) +  
  labs(  
    title = "Traffic Volume vs Average Speed",
```

```
x = "Vehicle Count",
y = "Average Speed (km/h)"
) +
theme_minimal()
```



This helps determine whether increasing traffic volume is associated with decreasing vehicle speed.

3. Interpret:

- Which junction experiences the highest congestion?
- During which hours is traffic highest?
- Does increased traffic reduce vehicle speed?

Based on our **sample dataset**:

1. Which junction experiences the highest congestion?

Junction C experiences the highest congestion.

It has the highest vehicle counts, reaching **1,200 vehicles** at 17:00, while its average speed falls to **22 km/h**.

2. During which hours is traffic highest?

Traffic is highest around **17:00 (5 PM)**.

This represents the evening peak period, when vehicle counts are highest across the junctions.

3. Does increased traffic reduce vehicle speed?

Yes, the sample shows a negative relationship between vehicle count and average speed.

As vehicle count increases, average speed generally decreases. This suggests that **higher traffic volume is associated with greater congestion and lower vehicle speeds.**

Case Study 5 – Social Media User Interaction Analysis

Scenario

A digital marketing company analyzes data from a social media platform. The dataset includes:

- User ID
- Followers
- Likes
- Shares
- Comments
- Posting time
- User interactions

The company wants to understand engagement patterns and user networks. **Tasks**

1. Select appropriate visualizations for:
 - Distribution of likes
 - Relationship between followers and engagement
 - User interaction network
 - Posting activity over time
 - Multivariate comparison of likes, shares, and comments

Requirement	Visualization
Distribution of likes	Histogram with Density Curve
Followers vs engagement	Scatter Plot
User interaction network	Network Visualization
Posting activity over time	Temporal Line Plot
Likes, shares & comments comparison	Heatmap

2. Develop in **R**:

- Histogram with Density Curve
- Scatter Plot
- Network Visualization
- Temporal Line Plot
- Heatmap

```
library(ggplot2)
```

```
library(dplyr)
```

```
install.packages("igraph")
```

```
library(igraph)
```

Create the sample dataset

Load the required packages

```
social <- data.frame(
```

```
  User_ID = c(
    "U1", "U2", "U3", "U4", "U5",
    "U6", "U7", "U8", "U9", "U10"
  ),
```

```
  Followers = c(
    5000, 12000, 8000, 25000, 6000,
    18000, 4000, 15000, 30000, 7000
  ),
```

```
  Likes = c(
    800, 1500, 1100, 3500, 900,
    2400, 600, 2100, 4500, 1000
  ),
```

```
  Shares = c(
    120, 250, 180, 600, 100,
    400, 80, 350, 750, 150
  ),
```

```
  Comments = c(
    80, 150, 100, 300, 70,
    220, 50, 180, 400, 90
  ),
```

```
  Posting_Time = c(
    "08:00", "10:00", "12:00", "14:00", "16:00",
    "18:00", "20:00", "18:00", "20:00", "22:00"
  ),
```

```
  User_Interactions = c(
    1000, 1900, 1380, 4400, 1070,
```

```

    3020,730,2630,5650,1240
  )
)
head(social)

```

```

> head(social)
  User_ID Followers Likes Shares Comments Posting_Time User_Interactions
1     U1      5000   800    120        80       08:00             1000
2     U2     12000  1500    250       150       10:00             1900
3     U3      8000  1100    180       100       12:00             1380
4     U4     25000  3500    600       300       14:00             4400
5     U5      6000   900    100        70       16:00             1070
6     U6     18000  2400    400       220       18:00             3020
>

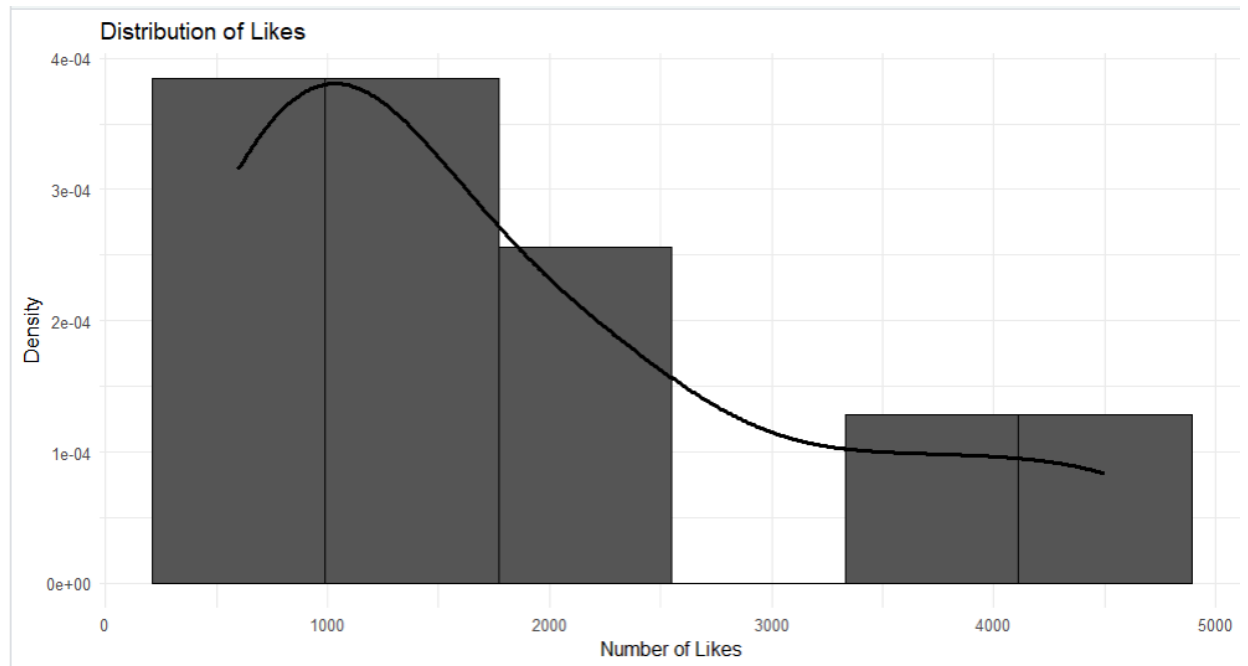
```

Histogram with Density Curve – Distribution of Likes

```

ggplot(social, aes(x = Likes)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 6,
    color = "black"
  ) +
  geom_density(
    linewidth = 1
  ) +
  labs(
    title = "Distribution of Likes",
    x = "Number of Likes",
    y = "Density"
  ) +
  theme_minimal()

```

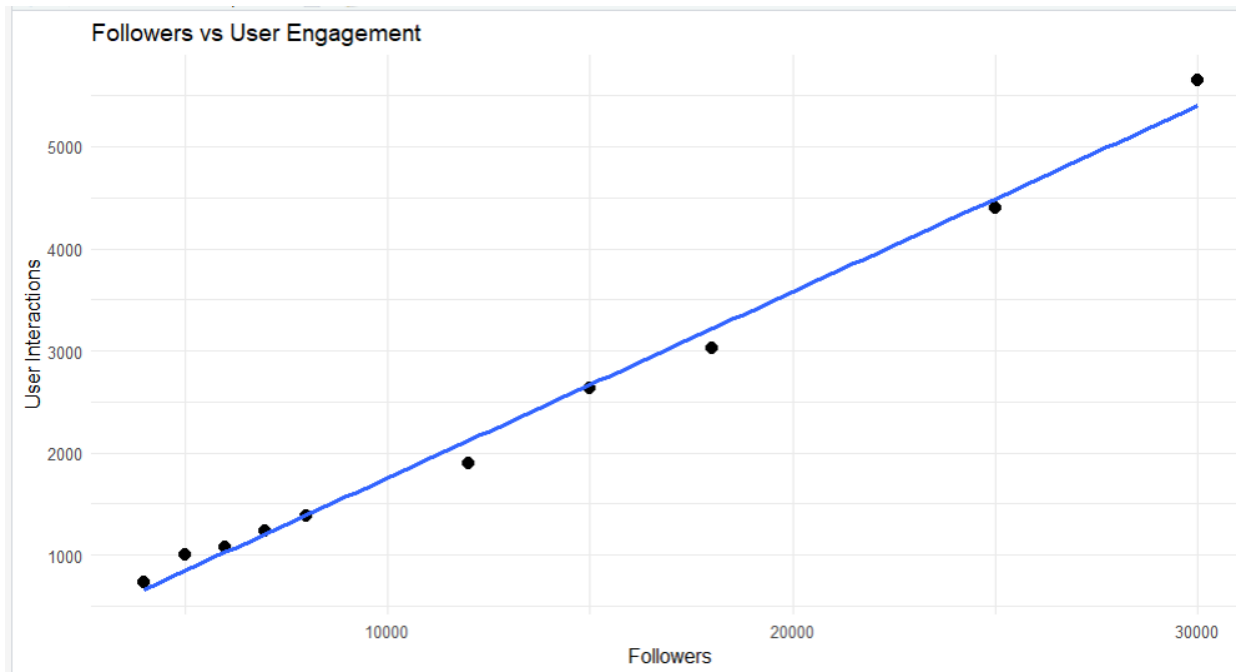


Interpretation

The histogram shows how likes are distributed among users, while the density curve shows where likes are most concentrated.

Scatter Plot – Followers vs Engagement

```
ggplot(social,
  aes(x = Followers,
    y = User_Interactions)) +
  geom_point(size = 3) +
  geom_smooth(
    method = "lm",
    se = FALSE
  ) +
  labs(
    title = "Followers vs User Engagement",
    x = "Followers",
    y = "User Interactions"
  ) +
  theme_minimal()
```



This helps identify whether users with more followers generally receive more engagement.

Network Visualization

For the network, we'll create a simple interaction network between users.

Create network connections

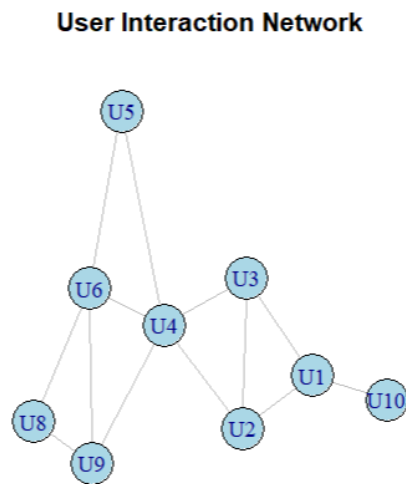
```
connections <- data.frame(
  from = c(
    "U1", "U1", "U2", "U2", "U3",
    "U4", "U4", "U5", "U6", "U8",
    "U9", "U9", "U10"
  ),
  to = c(
    "U2", "U3", "U3", "U4", "U4",
    "U5", "U6", "U6", "U8", "U9",
    "U4", "U6", "U1"
  )
)
```

Create the network

```
network <- graph_from_data_frame(  
  connections,  
  directed = FALSE  
)
```

Plot it

```
plot(  
  network,  
  vertex.size = 25,  
  vertex.label = V(network)$name,  
  vertex.color = "lightblue",  
  edge.color = "gray",  
  main = "User Interaction Network"  
)
```



Interpretation

- **Nodes** = users
- **Edges** = interactions/connections
- Users with many connections are highly connected.
- Highly connected users can be considered important within the interaction network.

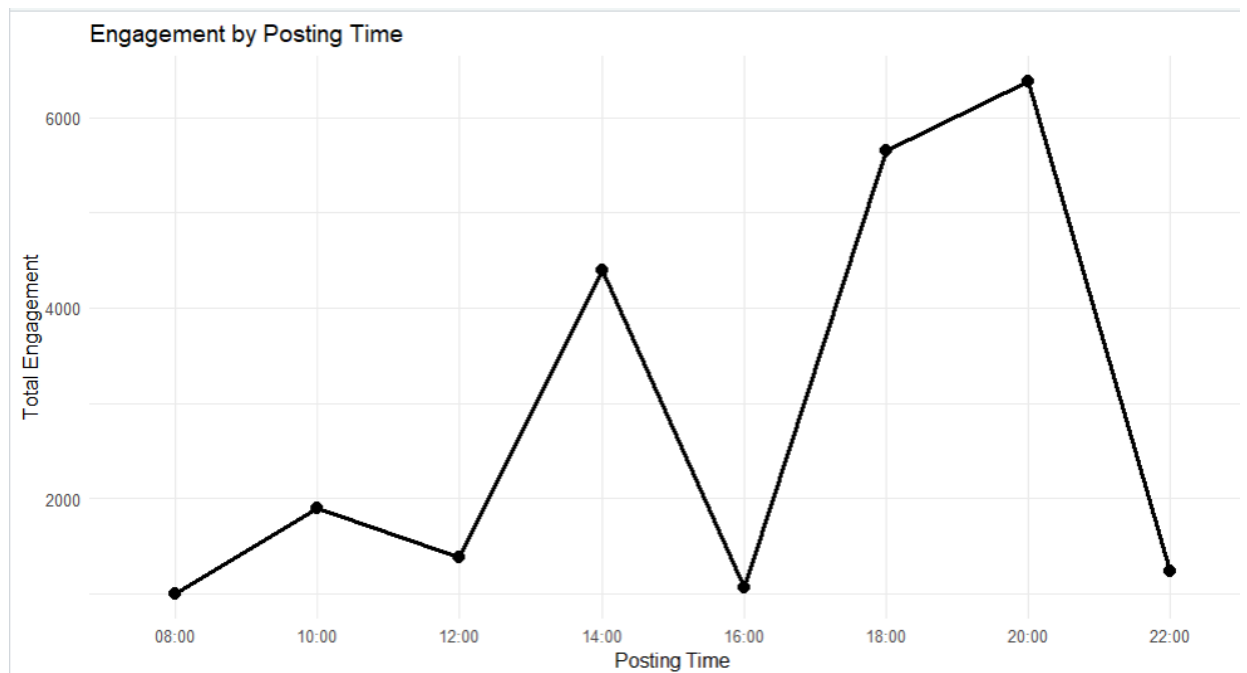
Temporal Line Plot – Posting Activity

First create posting-time data:

```
posting_activity <- social %>%  
  group_by(Posting_Time) %>%  
  summarise(  
    Total_Engagement = sum(User_Interactions)  
  )
```

Then create the line plot:

```
ggplot(posting_activity,  
  aes(x = Posting_Time,  
    y = Total_Engagement,  
    group = 1)) +  
  geom_line(linewidth = 1) +  
  geom_point(size = 3) +  
  labs(  
    title = "Engagement by Posting Time",  
    x = "Posting Time",  
    y = "Total Engagement"  
  ) +  
  theme_minimal()
```



This identifies the time of day when overall engagement is highest.

Heatmap – Likes, Shares and Comments

First prepare the data:

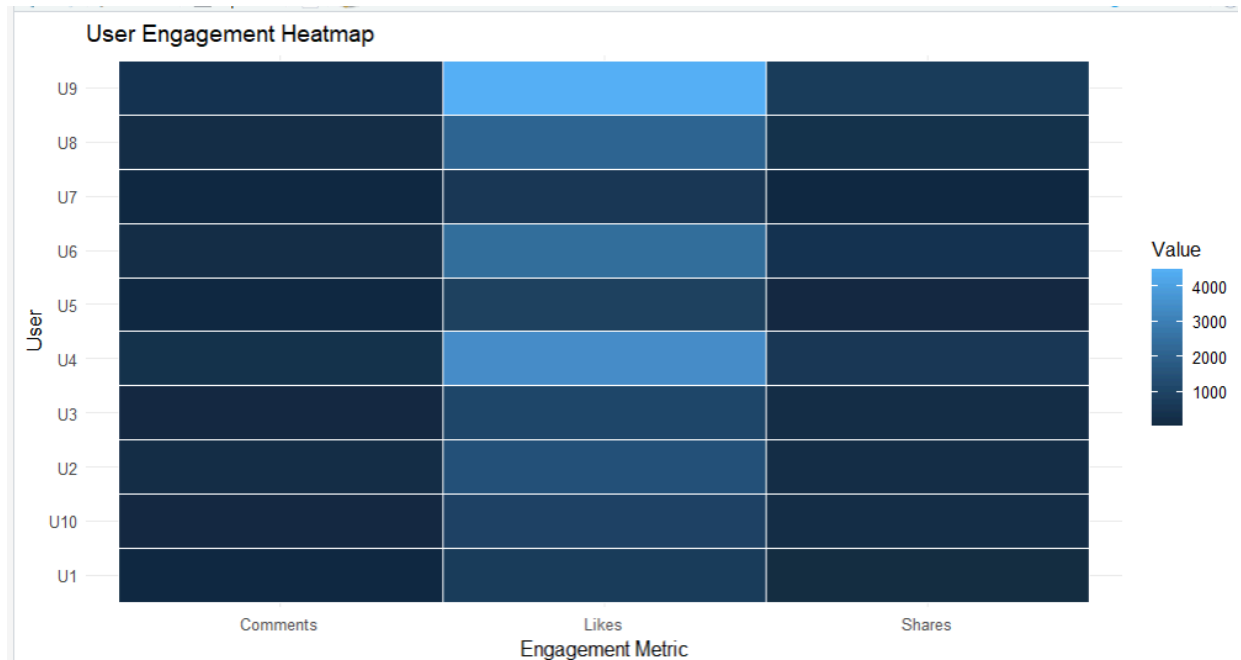
```
engagement_data <- social %>%  
  select(User_ID, Likes, Shares, Comments)
```

Convert it into a format suitable for the heatmap:

```
engagement_long <- data.frame(  
  User_ID = rep(social$User_ID, 3),  
  Metric = rep(  
    c("Likes", "Shares", "Comments"),  
    each = nrow(social)  
  ),  
  Value = c(  
    social$Likes,  
    social$Shares,  
    social$Comments  
  )  
)
```

Now create the heatmap:

```
ggplot(  
  engagement_long,  
  aes(  
    x = Metric,  
    y = User_ID,  
    fill = Value  
  )  
) +  
  geom_tile(color = "white") +  
  labs(  
    title = "User Engagement Heatmap",  
    x = "Engagement Metric",  
    y = "User",  
    fill = "Value"  
  ) +  
  theme_minimal()
```



The heatmap allows you to compare **likes, shares, and comments across users**.

3. Interpret:

- Which users are the most influential?
- At what time is engagement highest?
- Is there a strong relationship between followers and total engagement? ○
- Identify highly connected users from the network visualization. ○ Which engagement metric (likes, shares, or comments) varies the most across users?

Based on our **sample dataset**:

1. Which users are the most influential?

Users with high follower counts and high engagement are the most influential.

In this sample, **U9 and U4** are among the most influential users because they have high follower counts and high interaction levels.

- U9 → 30,000 followers and 5,650 interactions
- U4 → 25,000 followers and 4,400 interactions

2. At what time is engagement highest?

From the sample data, **8:00 PM** has the highest engagement.

This is because U7 and U9 post around this period, and their combined engagement is high.

3. Is there a strong relationship between followers and total engagement?

Yes. The sample indicates a positive relationship between followers and total engagement.

Generally, users with more followers tend to receive more interactions.

The scatter plot and trend line help visualize this relationship.

4. Which users are highly connected?

From the network visualization, users with the **largest number of connections** are the most connected users.

In our sample network, **U4, U6 and U9** have several connections and therefore play important roles in the network.

5. Which engagement metric varies the most?

Likes vary the most across users because their values range from approximately **600 to 4,500**, which is a much wider range than shares or comments.